

Elementary and complete symmetric polynomials

Elementary symmetric polynomials

For some arbitrarily large n , fix variables $[x_i]_{i=1}^n$

Then an Elementary symmetric polynomial of degree k is defined as

$$e_{\langle k, n \rangle} := \begin{cases} \sum_{\lambda \in \Lambda} \prod_{i=1}^n x_{\lambda_i} \left(\Lambda := \{ \langle \alpha_i \rangle_{i=1}^k \mid \forall i, j \leq k, i < j \iff \alpha_i < \alpha_j \} \right) & \text{iff } k \leq n \\ 0 & \text{otherwise} \end{cases}$$

Lemma

$$\prod_{i=1}^n (1 + t \cdot x_i) = \sum_{d=0}^n t^d \cdot e_{\langle d, n \rangle}$$

Complete or Homogeneous symmetric polynomials

$$h_d := \sum_{\alpha: \sum \alpha_i = d} x^\alpha = \sum_{\lambda \in \mathcal{P}(d, n)} m_\lambda(x)$$
$$h_0 := 1$$

Lemma

$$\prod_{i=1}^n \left(\frac{1}{1 - t \cdot x_i} \right) = \sum_{d \geq 0} t^d \cdot h_d(x)$$